

Dynamic PA power mode DPD on base of 2-dimensional Chebyshev polynomials

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Abstract—This paper addresses the challenge of power amplifier (PA) behavioral modeling in scenarios where the PA output power is dynamically varying and non-stationary. Since different PA power modes correspond to distinct nonlinearities, conventional behavioral models—such as the generalized memory polynomial (GMP) and Chebyshev polynomials—require extensions to account for these characteristics. In this work, a two-dimensional Chebyshev polynomial-based model is employed to capture nonlinear distortion across a wide range of PA output power levels, from 0.069 W to 0.912 W, and to predict nonlinear behavior for intermediate power modes. The proposed model achieves high performance, with an adjacent channel leakage ratio (ACLR) of less than -45 dB for the considered PA power modes. Furthermore, the model demonstrates up to a 14 dB improvement in performance compared to a model with the same number of parameters but without PA power mode information.

Index Terms—digital pre-distortion (DPD), Chebyshev polynomials, power amplifier dynamic mode, least squares method (LS), adjacent channel leakage ratio (ACLR)

I. INTRODUCTION

Power amplifiers (PAs) introduce nonlinearities that significantly affect the quality of radio frequency signals, degrading the performance of the transmitted signal. The distortions caused by PAs can be categorized as in-band and out-of-band interferences. Without a linearization method, these distortions increase the bit error rate (BER) at the receiver, generate interference signals, and degrade the transmission quality of adjacent-band signals. To mitigate these distortions, the digital pre-distortion (DPD) technique is widely used in modern base stations and cellular devices [1]–[3].

The DPD system is implemented as a block with an inverse nonlinear characteristic, modifying the PA input signal to minimize nonlinear distortion at the PA output (Fig. 1).

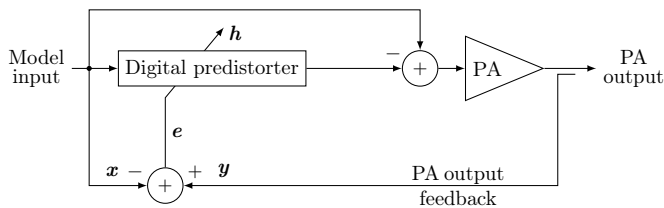


Fig. 1: DPD structure

Thus, proper implementation increases PA usage efficiency, due to the fact that it can be exploited in highly non-linear modes [1].

The digital pre-distortion (DPD) optimization task can be expressed by the following mathematical equation [4], which describes equalization of DPD input and PA output in block-wise mode:

$$\|PA(\mathbf{x} - \text{DPD}(\mathbf{x}, \mathbf{h})) - \mathbf{x}\|_2^2 \rightarrow \min_{\mathbf{h}}, \quad (1)$$

where $\mathbf{x} \in \mathbb{C}^{N \times 1}$ – PA input signal vector, $\mathbf{h} \in \mathbb{C}^{P \times 1}$ – DPD model parameters vector, N – block length. Consider that the PA nonlinear function $PA(\mathbf{x})$ can be linearized in the vicinity of the PA input:

$$PA(\mathbf{x} - \text{DPD}(\mathbf{x}, \mathbf{h})) \approx PA(\mathbf{x}) - (D_x PA(\mathbf{x}))\text{DPD}(\mathbf{x}, \mathbf{h}), \quad (2)$$

where $D_x PA(\mathbf{x}) \in \mathbb{C}^{N \times N}$ – PA output derivative w.r.t. the PA input. Since the DPD must operate in a PA mode where the level of nonlinear distortions remains significantly lower than the transmitter signal, we assume that $D_x PA(\mathbf{x}) \approx \mathbf{I}$, the identity matrix. Thus, substituting (2) into (1) we derive expression:

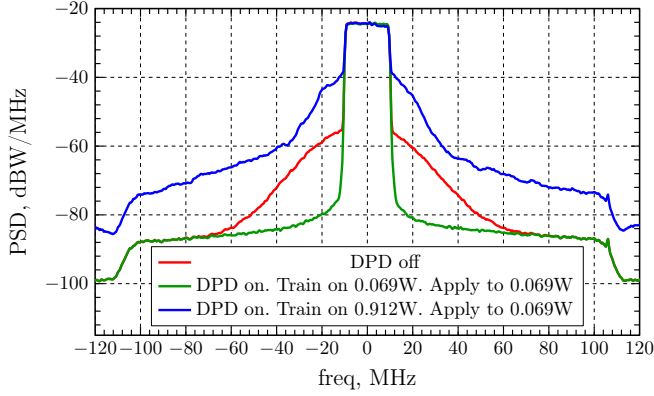
$$\|\text{DPD}(\mathbf{x}, \mathbf{h}) - \mathbf{e}\|_2^2 \rightarrow \min_{\mathbf{h}}, \quad (3)$$

where $\mathbf{e} = PA(\mathbf{x}) - \mathbf{x}$ – error vector. The optimization problem (3) is widely studied in papers on DPD [2]–[7]. This task is usually solved using least squares (LS), recursive least squares (RLS), and least mean squares (LMS) algorithms [8], [9], among others. In the proposed letter we exploit LS method for single-layer models training, because it achieves quadratic loss function global optimum in one optimization step.

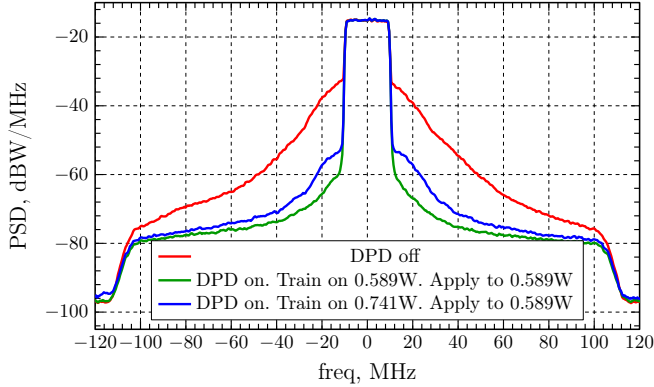
In the communication systems in order to meet certain requirements, resource blocks (RBs) in a data frame may be dynamically allocated according to real-time traffic, which can result in short time period power level changes thus reducing PA efficiency. This fact contradicts the assumption that DPD should operate in a stationary mode. This problem has been studied in [10].

On the other hand, it is necessary to significantly vary the input power levels of the PA in order to improve power efficiencies. To compensate the distortions caused by power changes DPD must be re-calibrated in real-time, which is not often economically justified in practice.

Moreover, transient processes during PA power mode switching result in distortions that propagate through the communication channel, leading to brief communication protocol violations and spectral mask degradation. This leads to a performance degradation when attempting to apply model parameters trained in one PA power mode to signals, corresponding to another mode, which is shown by power spectral densities (PSD) comparison in fig. 2.



(a) Apply 1D model. Train on 0.912 W case, test on 0.069 W case



(b) Apply 1D model. Train on 0.741 W case, test on 0.589 W case

Fig. 2: PSD of PA output. Dynamic PA mode non-stationarity illustration

In fig. 2a parameters of single-dimensional polynomial, trained on 0.069 W (green), 0.912 W (blue) cases are applied to 0.069 W case. One can observe up to 32 dB adjacent channel leakage ratio (ACLR) degradation because of different PA behavior for chosen power modes. However, when the model is trained on nearby PA output power modes, 0.598 W (green) and 0.714 W (blue) (fig. 2b), the performance degradation is 7.5 dB.

As a result, increased non-linear distortions may propagate through the communication channel while adjustment of the parameter. Consequently, a more favorable approach is to predict nonlinear distortions across all operating modes of the amplifier rather than track power modes.

To predict nonlinear distortions caused by various PA power modes, researchers primarily propose different approaches for

parameter recalculation relative to a chosen reference.

For instance, the paper [11] suggests dividing trainable model parameters into static and dynamic components, where dynamic parameters are switched according to the PA power mode. The work [3] extends the concept proposed in [11] by applying the power-adaptive decomposed vector rotation (PDVR) model.

Another notable approach involves AI-based methods [6], [7]. For example, the paper [6] proposes using an artificial neural network to gather information related to different nonlinearities and transform the general memory polynomial (GMP) model parameters to those corresponding to the target power, effectively predicting PA nonlinearity for a given power mode.

In this article, we introduce an alternative dynamic PA power compensation method based on two-dimensional Chebyshev polynomial offline training. In fact it represents traditional approach, which takes into consideration dynamic conditions within 2-nd dimension of non-linear model.

Note that general view of optimization problem, related to dynamic PA power scenario might be represented as the sum of loss functions corresponding to each power case (3):

$$\sum_{i=0}^{C-1} \|\text{DPD}(\mathbf{x}, \mathbf{p}, \mathbf{h}) - \mathbf{e}_i\|_2^2 \rightarrow \min_{\mathbf{h}}, \quad (4)$$

where C – number of PA power cases, $\mathbf{p} \in \mathbb{C}^{N \times 1}$ – vector of PA input power features. Thus, $(\mathbf{x}, \mathbf{p})$ – 2-dimensional model input, $\mathbf{h} \in \mathbb{C}^{P \times 1}$ – DPD model parameters vector.

II. 2D CHEBYSHEV POLYNOMIAL DPD TECHNIQUE

Among non-linear digital signal processing tasks polynomials [1], [3], spline-based polynomials [12] are often used for the purpose of signal non-linearity description. Common single-dimensional DPD based on Chebyshev polynomial for one-dimensional input signal is represented as:

$$z_n = \sum_{i=0}^{P-1} h_i x_n T_i(|x_n|), \quad (5)$$

$$T_i(|x_n|) = \cos(i \arccos(|x_n|)),$$

here P – Chebyshev polynomial order, $\{h_i\}_{i=0}^{P-1}$ – Chebyshev polynomial trainable parameters.

Non-linear model is required to take into account non-linear memory of the PA. For this reason proposed model (6) is expanded to the sum of non-linearities, fed by delayed signals $|x_{n-d}|$, where d – signal delay corresponding to the d -th branch. Thus, model output is shown as follows:

$$y_n = \sum_{d=-D}^D \sum_{i=0}^{P-1} h_{k,i} x_{n-d} T_i(|x_{n-d}|), \quad (6)$$

where $B = 2D + 1$ – number of model branches. Fig. 3 shows general scheme of chosen parallel non-linear Chebyshev polynomial-based model.

In current paper single-dimensional Chebyshev polynomial is tested on the case of dynamically changing PA output

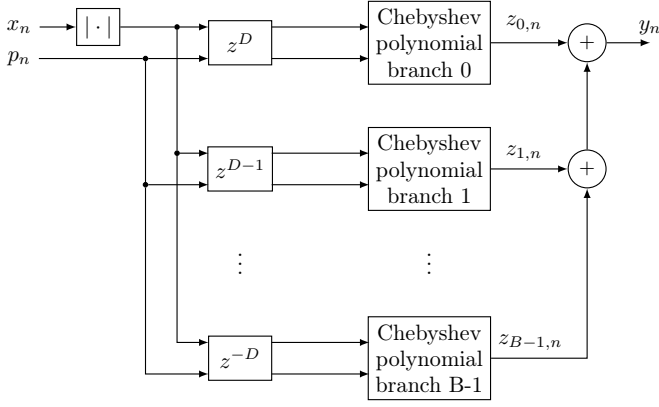


Fig. 3: B -branch non-linear Chebyshev polynomial-based model

power. According to the results, represented in section IV, 1-dimensional (1D) non-linearity can't absorb dynamically changing PA power properties. Therefore, current article introduces 2-dimensional (2D) non-linearity, which is fed by both features: signal magnitude $|x_n|$ and feature, which corresponds to current PA power mode p_n . The given feature p_n is PA input power, which uniformly increases in linear scale. Totally, $C = 61$ input power cases are considered:

$$25.1 \mu\text{W}, 37.9 \mu\text{W}, 50.8 \mu\text{W}, \dots, 794.3 \mu\text{W} \quad (7)$$

Thus, 2-dimensional polynomial output is as following:

$$\begin{aligned} z_n &= \sum_{i=0}^{P_1-1} \sum_{j=0}^{P_2-1} h_{i,j} x_n T_{i,j}(|x_n|, p_n), \\ T_{i,j}(|x_n|, p_n) &= T_i(|x_n|) T_j(p_n) = \\ &= \cos(i \arccos(|x_n|)) \cos(j \arccos(p_n)), \end{aligned} \quad (8)$$

here P_1, P_2 – 2-dimensional Chebyshev polynomial orders related to magnitude and power mode feature dimensions correspondingly, $\{h_{i,j}\}_{i=0, j=0}^{P_1-1, P_2-1}$ – 2-dimensional Chebyshev polynomial trainable parameters. 2D non-linearity basis functions $T_{i,j}(\cdot)$ are expressed through the multiplication of 1D non-linearity basis functions $T_i(\cdot), T_j(\cdot)$, related to input signal magnitude and PA power correspondingly. Multi-branch 2-dimensional model output is represented as:

$$y_n = \sum_{d=-D}^D \sum_{i=0}^{P_1-1} \sum_{j=0}^{P_2-1} h_{i,j,k} x_{n-d} T_{i,j}(|x_{n-d}|, p_{n-d}), \quad (9)$$

where basis function $T_{i,j}$ is expressed similarly to (8).

Note, that input signal magnitude $|x_n|$ and PA mode feature p_n are implied to be scaled into the same range $[0, 1]$ for the purpose of data standardization and from the perspective of satisfying Chebyshev polynomial orthogonality conditions.

III. TESTBENCH DESCRIPTION

Testbench structure is shown in fig. 4. The measurement setup consists of personal computer (PC), signal generator

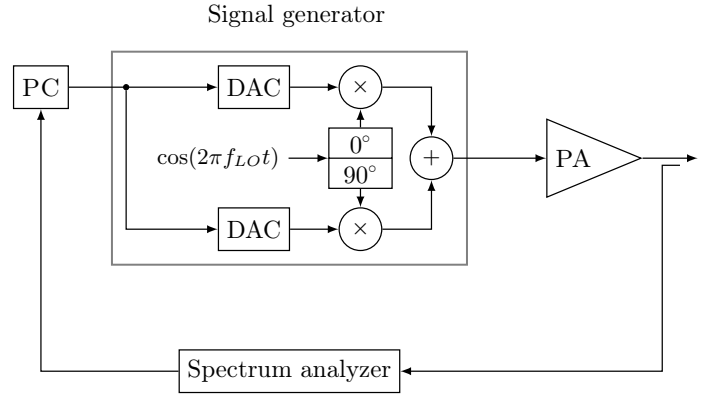


Fig. 4: The scheme of testbench

(SG) R&S SMW200A, spectrum analyzer (SA) R&S FSW85 and power amplifier ZKY66291-11 under testing.

PC loads baseband (BB) in-phase and quadrature components (IQ data) to SG which modulates BB to the carrier frequency and sends to the PA input. PA output with nonlinear distortions transmitted to SA. After that IQ BB data is sent to the PC for further signal processing.

Power control was done by SG. Complex valued 20 MHz orthogonal frequency-division multiplexing (OFDM) signal with 100 resource blocks, carrier $f_{LO} = 1.8$ GHz was used for experiments.

PA input power cases (7) correspond to $C = 61$ PA output power cases in dynamic range 11.2 dB:

$$\begin{aligned} 0.069 \text{ W}, 0.107 \text{ W}, 0.143 \text{ W}, \dots, 0.912 \text{ W} \\ -11.6 \text{ dBW}, -9.7 \text{ dBW}, -8.4 \text{ dBW}, \dots, -0.4 \text{ dBW} \end{aligned} \quad (10)$$

Each PA power case consists of 147k complex samples. Even power cases are used for training:

$$\begin{aligned} 0.069 \text{ W}, 0.143 \text{ W}, \dots, 0.912 \text{ W} \\ -11.6 \text{ dBW}, -8.4 \text{ dBW}, \dots, -0.4 \text{ dBW} \end{aligned} \quad (11)$$

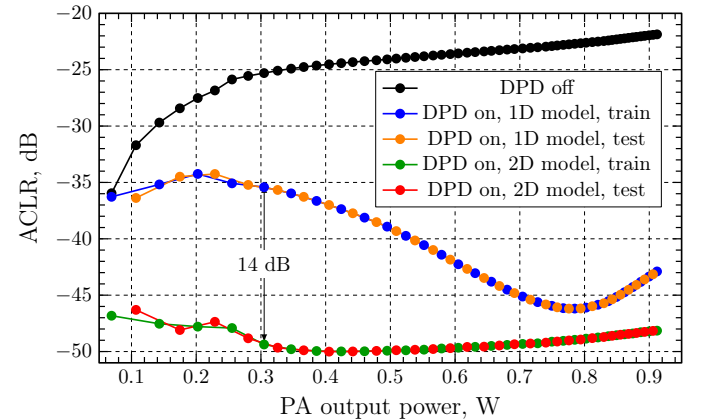
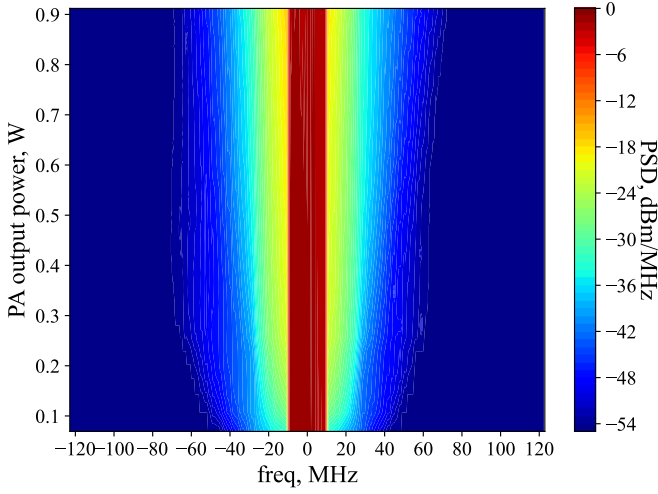
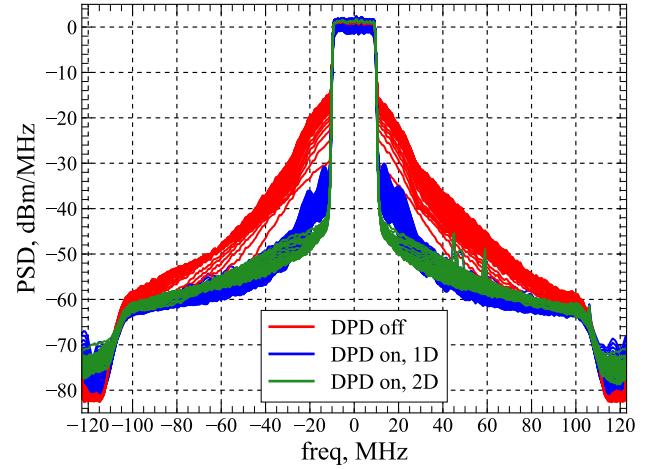


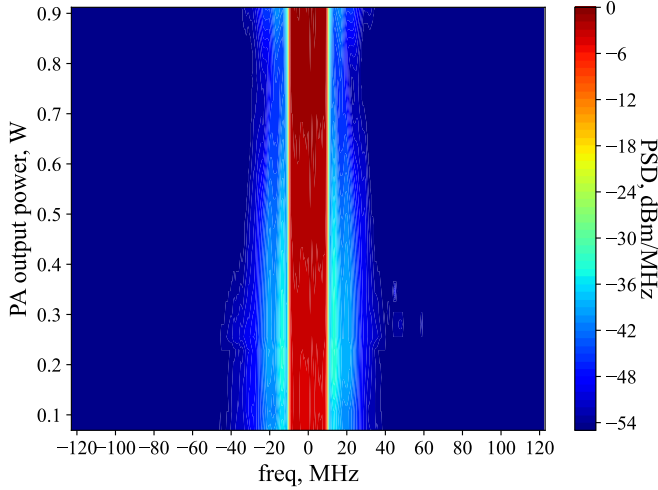
Fig. 5: Performance of 1D- and 2D-dimensional Chebyshev polynomial on train and test datasets



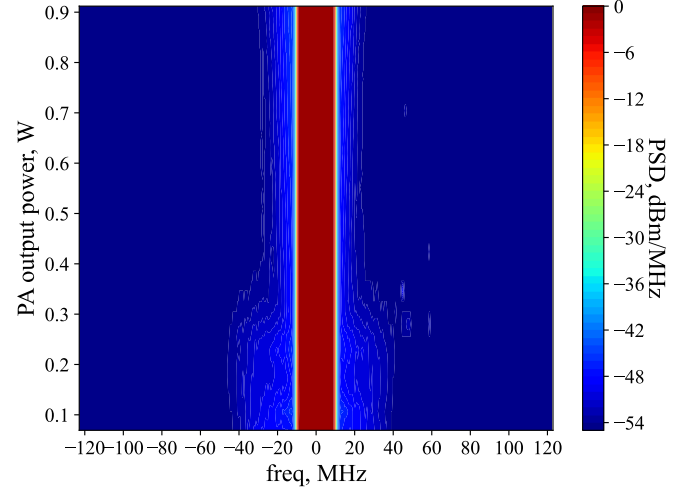
(a) DPD off



(b) DPD off/on, 1D and 2D Chebyshev polynomials signals for all considered cases



(c) DPD on, 1D Chebyshev polynomial



(d) DPD on, 2D Chebyshev polynomial

Fig. 6: PSD of PA output. Application results of DPD on base of 1D and 2D Chebyshev polynomials

Odd power cases are considered for model performance evaluation:

$$\begin{aligned} &0.107 \text{ W}, 0.175 \text{ W}, \dots, 0.906 \text{ W} \\ &-9.7 \text{ dBW}, -7.6 \text{ dBW}, \dots, -0.4 \text{ dBW}. \end{aligned} \quad (12)$$

The source code and experimental data used in this study are available at the GitHub repository [13].

IV. EXPERIMENTAL RESULTS

The number of branches in two-dimensional model $B = 9$. Thus, delays are in range $[-4, \dots, 4]$, which is enough to describe PA memory effects according to simulations results. The orders of Chebyshev polynomial along signal magnitude $|x_n|$ and PA power feature p_n dimensions $P_1 = 22$ and $P_2 = 10$ correspondingly.

Chebyshev polynomial order P_2 is chosen to achieve acceptable ACLR values -45 dB in the whole range of powers.

Total number of parameters for single-dimensional model is chosen to equalize total number of parameters for both considered models.

Thus, 1D and 2D polynomials-based models include the same number of trainable complex parameters equal 1980.

Performance of both mentioned models is shown in fig. 5. According to the simulations results 2-dimensional model provides $\text{ACLR} < -45$ dB in the whole range of PA output powers. Moreover, there is an improvement up to 14 dB comparing to 1-dimensional polynomial among all considered PA power modes.

Figure 6 shows PSD of transmitted signal before DPD-based correction and after application of pre-trained 1D and 2D Chebyshev polynomials. Model evaluation is made both on train and test data, resulting corrected PA output PSD plots are visualized in fig. 6.

Fig. 6a shows transmitted signal PSD color map before DPD application in correspondence to all PA output powers cases.

Whereas fig. 6c, 6d provide PSD color map of transmitted signal after application of DPD model based on 1D and 2D Chebyshev polynomials correspondingly. In addition, fig. 6b represents PSD of TX before and after DPD application in the same plot with normalized output powers for different cases. Thus, according to fig. 5, 6c, 6d 2D-based DPD model provides significant performance improvement in comparison to the 1D-based model up to 14 dB among the whole chosen PA output power range: 0.069 W, $\dots$, 0.912 W.

V. CONCLUSION

Suggested offline DPD model, based on 2-dimensional Chebyshev polynomials, which provides PA output signal correction within the wide range of PA power modes: from 0.069 W to 0.912 W PA output power. The model is fed by PA input signal and the feature, which corresponds to PA input power. The model implemented and compared with DPD correction on base of 1-dimensional Chebyshev polynomials fed by PA input signal.

2D-based model provides ACLR < -45 dB performance for the whole range of considered PA output power cases.

In addition, suggested model provides ACLR improvement up to 14 dB for different PA output power cases in comparison with 1D Chebyshev polynomial-based model with the same number of trainable complex parameters.

Taking the aforementioned into account, 2D Chebyshev polynomial-based model enables prediction of PA behavior in response to changes in power mode and addresses the issue of non-stationarity during DPD parameter extraction.

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